PRAKTIKUM BASIS DATA

MODUL 3

Scripting_SQL_Basic

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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TP

1. **Soal 1**

   Script:

   ```
   CREATE TABLE mahasiswa (
      id NUMBER PRIMARY KEY,
      nama VARCHAR2(100),
      tempat_lahir VARCHAR2(50),
      tanggal_lahir DATE,
      nomor_hp VARCHAR2(15),
      email VARCHAR2(100),
      tinggi_badan NUMBER,
      berat_badan NUMBER
   );
   ```

   ```
   SQL> desc mahasiswa;
    Name                                      Null?    Type
    ----------------------------------------- -------- ----------------------
    ID                                        NOT NULL NUMBER
    NAMA                                               VARCHAR2(100)
    TEMPAT_LAHIR                                       VARCHAR2(50)
    TANGGAL_LAHIR                                      DATE
    NOMOR_HP                                           VARCHAR2(15)
    EMAIL                                              VARCHAR2(100)
    TINGGI_BADAN                                       NUMBER
    BERAT_BADAN                                        NUMBER
   ```

2. **Soal 2**

ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHI	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	Socrates	Athens	04-JUN-00	081234560001	socrates@uni.com	170	65
2	Plato	Athens	10-FEB-01	081234560002	plato@uni.com	168	63
3	Aristotle	Stagira	12-MAY-00	081234560003	aristotle@uni.com	172	67
4	Friedrich Nietzsche	Rocken	15-OCT-01	081234560004	nietzsche@uni.com	175	70
5	Albert Camus	Mondovi	07-NOV-02	081234560005	camus@uni.com	173	68
6	Marcus Aurelius	Rome	21-APR-01	081234560006	aurelius@uni.com	178	74
7	Leonardo da Vinci	Vinci	30-MAR-02	081234560007	davinci@uni.com	176	72
8	Pablo Escobar	Medellin	11-JAN-01	081234560008	escobar@uni.com	174	80
9	Geralt of Rivia	Rivia	09-AUG-00	081234560009	geralt@uni.com	180	78
10	Arthur Morgan	New Hanover	17-DEC-01	081234560010	arthur@uni.com	182	85
11	Ezio Auditore	Florence	25-JUN-02	081234560011	ezio@uni.com	177	73
12	Kratos	Sparta	13-SEP-00	081234560012	kratos@uni.com	190	95
13	Lara Croft	London	19-FEB-02	081234560013	lara@uni.com	167	55
14	Ada Wong	Unknown	05-APR-01	081234560014	ada@uni.com	165	53
15	Link	Hyrule	28-JUL-02	081234560015	link@uni.com	169	60

3. **Soal 3**

   a. **Karakter dan String**

      - **Lower/Upper**

        Script:

```
SELECT nama,
    LOWER(nama) AS huruf_kecil,
    UPPER(nama) AS huruf_besar
FROM mahasiswa;
```

NAMA	HURUF_KECIL	HURUF_BESAR
Socrates	socrates	SOCRATES
Plato	plato	PLATO
Aristotle	aristotle	ARISTOTLE
Friedrich Nietzsche	friedrich nietzsche	FRIEDRICH NIETZSCHE
Albert Camus	albert camus	ALBERT CAMUS
Marcus Aurelius	marcus aurelius	MARCUS AURELIUS
Leonardo da Vinci	leonardo da vinci	LEONARDO DA VINCI
Pablo Escobar	pablo escobar	PABLO ESCOBAR
Geralt of Rivia	geralt of rivia	GERALT OF RIVIA
Arthur Morgan	arthur morgan	ARTHUR MORGAN
Ezio Auditore	ezio auditore	EZIO AUDITORE
Kratos	kratos	KRATOS
Lara Croft	lara croft	LARA CROFT
Ada Wong	ada wong	ADA WONG
Link	link	LINK

- **INITCAP**

  Script:

```
SELECT nama,
    INITCAP(nama) AS kapital_awal
FROM mahasiswa;
```

NAMA	KAPITAL_AWAL
Socrates	Socrates
Plato	Plato
Aristotle	Aristotle
Friedrich Nietzsche	Friedrich Nietzsche
Albert Camus	Albert Camus
Marcus Aurelius	Marcus Aurelius
Leonardo da Vinci	Leonardo Da Vinci
Pablo Escobar	Pablo Escobar
Geralt of Rivia	Geralt Of Rivia
Arthur Morgan	Arthur Morgan
Ezio Auditore	Ezio Auditore
Kratos	Kratos
Lara Croft	Lara Croft
Ada Wong	Ada Wong
Link	Link

- **CONCAT**

  Script:

  ```
  SELECT nama,
      CONCAT(nama, ' - Mahasiswa') AS keterangan
  FROM mahasiswa;
  ```

  ```
  NAMA                 KETERANGAN
  ___________________  ____________________________________
  Socrates             Socrates - Mahasiswa
  Plato                Plato - Mahasiswa
  Aristotle            Aristotle - Mahasiswa
  Friedrich Nietzsche  Friedrich Nietzsche - Mahasiswa
  Albert Camus         Albert Camus - Mahasiswa
  Marcus Aurelius      Marcus Aurelius - Mahasiswa
  Leonardo da Vinci    Leonardo da Vinci - Mahasiswa
  Pablo Escobar        Pablo Escobar - Mahasiswa
  Geralt of Rivia      Geralt of Rivia - Mahasiswa
  Arthur Morgan        Arthur Morgan - Mahasiswa
  Ezio Auditore        Ezio Auditore - Mahasiswa
  Kratos               Kratos - Mahasiswa
  Lara Croft           Lara Croft - Mahasiswa
  Ada Wong             Ada Wong - Mahasiswa
  Link                 Link - Mahasiswa
  ```

- **LENGTH**

  Script:

  ```
  SELECT nama,
      LENGTH(nama) AS panjang_nama
  FROM mahasiswa;
  ```

NAMA	PANJANG_NAMA
Socrates	8
Plato	5
Aristotle	9
Friedrich Nietzsche	19
Albert Camus	12
Marcus Aurelius	15
Leonardo da Vinci	17
Pablo Escobar	13
Geralt of Rivia	15
Arthur Morgan	13
Ezio Auditore	13
Kratos	6
Lara Croft	10
Ada Wong	8
Link	4

- **SUBSTR**

  Script:

  SELECT nama,

  SUBSTR(nama,1,4) AS potongan_nama

  FROM mahasiswa;

NAMA	POTONGAN_NAMA
Socrates	Socr
Plato	Plat
Aristotle	Aris
Friedrich Nietzsche	Frie
Albert Camus	Albe
Marcus Aurelius	Marc
Leonardo da Vinci	Leon
Pablo Escobar	Pabl
Geralt of Rivia	Gera
Arthur Morgan	Arth
Ezio Auditore	Ezio
Kratos	Krat
Lara Croft	Lara
Ada Wong	Ada
Link	Link

- **LPAD/RPAD**

  Script:

  SELECT nama,

  LPAD(nama,20,'*') AS kiri,

  RPAD(nama,20,'-') AS kanan

  FROM mahasiswa;

NAMA	KIRI	KANAN
Socrates	************Socrates	Socrates————————
Plato	***************Plato	Plato———————
Aristotle	***********Aristotle	Aristotle————————
Friedrich Nietzsche	*Friedrich Nietzsche	Friedrich Nietzsche-
Albert Camus	********Albert Camus	Albert Camus————
Marcus Aurelius	*****Marcus Aurelius	Marcus Aurelius——
Leonardo da Vinci	**Leonardo da Vinci	Leonardo da Vinci——
Pablo Escobar	*******Pablo Escobar	Pablo Escobar————
Geralt of Rivia	*****Geralt of Rivia	Geralt of Rivia——
Arthur Morgan	*******Arthur Morgan	Arthur Morgan————
Ezio Auditore	*******Ezio Auditore	Ezio Auditore————
Kratos	**************Kratos	Kratos———————
Lara Croft	**********Lara Croft	Lara Croft—————
Ada Wong	************Ada Wong	Ada Wong—————
Link	***************Link	Link———————

- **LTRIM/RTRIM**

  Script:

  SELECT LTRIM(' '||nama) AS hapus_spasi_kiri,

  RTRIM(nama||' ') AS hapus_spasi_kanan

  FROM mahasiswa;

HAPUS_SPASI_KIRI	HAPUS_SPASI_KANAN
Socrates	Socrates
Plato	Plato
Aristotle	Aristotle
Friedrich Nietzsche	Friedrich Nietzsche
Albert Camus	Albert Camus
Marcus Aurelius	Marcus Aurelius
Leonardo da Vinci	Leonardo da Vinci
Pablo Escobar	Pablo Escobar
Geralt of Rivia	Geralt of Rivia
Arthur Morgan	Arthur Morgan
Ezio Auditore	Ezio Auditore
Kratos	Kratos
Lara Croft	Lara Croft
Ada Wong	Ada Wong
Link	Link

- **INSTR**

  Script:

  SELECT email,

```
    INSTR(email,'@') AS posisi_at
FROM mahasiswa;
```

EMAIL	POSISI_AT
socrates@uni.com	9
plato@uni.com	6
aristotle@uni.com	10
nietzsche@uni.com	10
camus@uni.com	6
aurelius@uni.com	9
davinci@uni.com	8
escobar@uni.com	8
geralt@uni.com	7
arthur@uni.com	7
ezio@uni.com	5
kratos@uni.com	7
lara@uni.com	5
ada@uni.com	4
link@uni.com	5

- **REPLACE**

  Script:

```
SELECT email,
    REPLACE(email,'uni','kampus') AS email_baru
FROM mahasiswa;
```

EMAIL	EMAIL_BARU
socrates@uni.com	socrates@kampus.com
plato@uni.com	plato@kampus.com
aristotle@uni.com	aristotle@kampus.com
nietzsche@uni.com	nietzsche@kampus.com
camus@uni.com	camus@kampus.com
aurelius@uni.com	aurelius@kampus.com
davinci@uni.com	davinci@kampus.com
escobar@uni.com	escobar@kampus.com
geralt@uni.com	geralt@kampus.com
arthur@uni.com	arthur@kampus.com
ezio@uni.com	ezio@kampus.com
kratos@uni.com	kratos@kampus.com
lara@uni.com	lara@kampus.com
ada@uni.com	ada@kampus.com
link@uni.com	link@kampus.com

b. **Tanggal dan Waktu**

- **Format tanggal (MM, MON, MONTH)**

  Script:

```
SELECT tanggal_lahir,
    TO_CHAR(tanggal_lahir,'MM') AS bulan_angka,
    TO_CHAR(tanggal_lahir,'MON') AS bulan_singkat,
    TO_CHAR(tanggal_lahir,'MONTH') AS bulan_panjang
FROM mahasiswa;
```

TANGGAL_LAHI	BU	BULAN_SINGKA	BULAN_PANJANG
04-JUN-00	06	JUN	JUNE
10-FEB-01	02	FEB	FEBRUARY
12-MAY-00	05	MAY	MAY
15-OCT-01	10	OCT	OCTOBER
07-NOV-02	11	NOV	NOVEMBER
21-APR-01	04	APR	APRIL
30-MAR-02	03	MAR	MARCH
11-JAN-01	01	JAN	JANUARY
09-AUG-00	08	AUG	AUGUST
17-DEC-01	12	DEC	DECEMBER
25-JUN-02	06	JUN	JUNE
13-SEP-00	09	SEP	SEPTEMBER
19-FEB-02	02	FEB	FEBRUARY
05-APR-01	04	APR	APRIL
28-JUL-02	07	JUL	JULY

- **DD/DY**

  Script:

```
SELECT tanggal_lahir,
    TO_CHAR(tanggal_lahir,'DD') AS hari,
    TO_CHAR(tanggal_lahir,'DY') AS hari_singkat
FROM mahasiswa;
```

```
TANGGAL_LAHI HA HARI_SINGKAT
------------ -- ------------
04-JUN-00    04 SUN
10-FEB-01    10 SAT
12-MAY-00    12 FRI
15-OCT-01    15 MON
07-NOV-02    07 THU
21-APR-01    21 SAT
30-MAR-02    30 SAT
11-JAN-01    11 THU
09-AUG-00    09 WED
17-DEC-01    17 MON
25-JUN-02    25 TUE
13-SEP-00    13 WED
19-FEB-02    19 TUE
05-APR-01    05 THU
28-JUL-02    28 SUN
```

- **YYYY/ YY / RR**

  Script:

  SELECT tanggal_lahir,
  TO_CHAR(tanggal_lahir,'YYYY') AS tahun_full,
  TO_CHAR(tanggal_lahir,'YY') AS tahun_pendek
  FROM mahasiswa;

```
TANGGAL_LAHI TAHU TA
------------ ---- --
04-JUN-00    2000 00
10-FEB-01    2001 01
12-MAY-00    2000 00
15-OCT-01    2001 01
07-NOV-02    2002 02
21-APR-01    2001 01
30-MAR-02    2002 02
11-JAN-01    2001 01
09-AUG-00    2000 00
17-DEC-01    2001 01
25-JUN-02    2002 02
13-SEP-00    2000 00
19-FEB-02    2002 02
05-APR-01    2001 01
28-JUL-02    2002 02
```

- **AM / PM dan Jam**

  Script:

  SELECT SYSDATE,
  TO_CHAR(SYSDATE,'HH24:MI:SS') AS format_24,
  TO_CHAR(SYSDATE,'HH:MI:SS AM') AS format_12
  FROM dual;

SYSDATE	FORMAT_2	FORMAT_12
10-MAR-26	19:48:14	07:48:14 PM

- **ADD_MONTHS**

  Script:

  SELECT nama,
  tanggal_lahir,
  ADD_MONTHS(tanggal_lahir,6) AS tambah_6_bulan
  FROM mahasiswa;

NAMA	TANGGAL_LAHI	TAMBAH_6_
Socrates	04-JUN-00	04-DEC-00
Plato	10-FEB-01	10-AUG-01
Aristotle	12-MAY-00	12-NOV-00
Friedrich Nietzsche	15-OCT-01	15-APR-02
Albert Camus	07-NOV-02	07-MAY-03
Marcus Aurelius	21-APR-01	21-OCT-01
Leonardo da Vinci	30-MAR-02	30-SEP-02
Pablo Escobar	11-JAN-01	11-JUL-01
Geralt of Rivia	09-AUG-00	09-FEB-01
Arthur Morgan	17-DEC-01	17-JUN-02
Ezio Auditore	25-JUN-02	25-DEC-02
Kratos	13-SEP-00	13-MAR-01
Lara Croft	19-FEB-02	19-AUG-02
Ada Wong	05-APR-01	05-OCT-01
Link	28-JUL-02	28-JAN-03

- **SYSDATE**

  Script:

  SELECT SYSDATE FROM dual;

SYSDATE
10-MAR-26

- **LAST DAY**

  Script:

  ```
  SELECT tanggal_lahir,
      LAST_DAY(tanggal_lahir) AS akhir_bulan
  FROM mahasiswa;
  ```

TANGGAL_LAHI	AKHIR_BUL
04-JUN-00	30-JUN-00
10-FEB-01	28-FEB-01
12-MAY-00	31-MAY-00
15-OCT-01	31-OCT-01
07-NOV-02	30-NOV-02
21-APR-01	30-APR-01
30-MAR-02	31-MAR-02
11-JAN-01	31-JAN-01
09-AUG-00	31-AUG-00
17-DEC-01	31-DEC-01
25-JUN-02	30-JUN-02
13-SEP-00	30-SEP-00
19-FEB-02	28-FEB-02
05-APR-01	30-APR-01
28-JUL-02	31-JUL-02

- **NEXT DAY**

  Script:

  ```
  SELECT tanggal_lahir,
      NEXT_DAY(tanggal_lahir,'MONDAY') AS senin_berikutnya
  FROM mahasiswa;
  ```

TANGGAL_LAHI	SENIN_BER
04-JUN-00	05-JUN-00
10-FEB-01	12-FEB-01
12-MAY-00	15-MAY-00
15-OCT-01	22-OCT-01
07-NOV-02	11-NOV-02
21-APR-01	23-APR-01
30-MAR-02	01-APR-02
11-JAN-01	15-JAN-01
09-AUG-00	14-AUG-00
17-DEC-01	24-DEC-01
25-JUN-02	01-JUL-02
13-SEP-00	18-SEP-00
19-FEB-02	25-FEB-02
05-APR-01	09-APR-01
28-JUL-02	29-JUL-02

- **MONTHS BETWEEN**

  Script:

  SELECT nama,
  MONTHS_BETWEEN(SYSDATE,tanggal_lahir) AS selisih_bulan
  FROM mahasiswa;

NAMA	SELISIH_BULAN
Socrates	309.220255
Plato	301
Aristotle	309.96219
Friedrich Nietzsche	292.865416
Albert Camus	280.12348
Marcus Aurelius	298.671868
Leonardo da Vinci	287.381545
Pablo Escobar	301.994448
Geralt of Rivia	307.058964
Arthur Morgan	290.8009
Ezio Auditore	284.542835
Kratos	305.929932
Lara Croft	288.736384
Ada Wong	299.187997
Link	283.446061

- **ROUND tanggal**

  Script:

  SELECT tanggal_lahir,
  ROUND(tanggal_lahir,'YEAR') AS pembulatan_tahun
  FROM mahasiswa;

TANGGAL_LAHI	PEMBULATA
04-JUN-00	01-JAN-00
10-FEB-01	01-JAN-01
12-MAY-00	01-JAN-00
15-OCT-01	01-JAN-02
07-NOV-02	01-JAN-03
21-APR-01	01-JAN-01
30-MAR-02	01-JAN-02
11-JAN-01	01-JAN-01
09-AUG-00	01-JAN-01
17-DEC-01	01-JAN-02
25-JUN-02	01-JAN-02
13-SEP-00	01-JAN-01
19-FEB-02	01-JAN-02
05-APR-01	01-JAN-01
28-JUL-02	01-JAN-03

c. **Fungsi Konversi**

- **TO_NUMBER**

  Script:

  SELECT TO_NUMBER('150') AS contoh_number
  FROM dual;

CONTOH_NUMBER
150

- **TO_CHAR (date)**

  Script:

  SELECT nama,
  TO_CHAR(tanggal_lahir,'DD MONTH YYYY') AS tanggal_format
  FROM mahasiswa;

NAMA	TANGGAL_FORMAT
Socrates	04 JUNE 2000
Plato	10 FEBRUARY 2001
Aristotle	12 MAY 2000
Friedrich Nietzsche	15 OCTOBER 2001
Albert Camus	07 NOVEMBER 2002
Marcus Aurelius	21 APRIL 2001
Leonardo da Vinci	30 MARCH 2002
Pablo Escobar	11 JANUARY 2001
Geralt of Rivia	09 AUGUST 2000
Arthur Morgan	17 DECEMBER 2001
Ezio Auditore	25 JUNE 2002
Kratos	13 SEPTEMBER 2000
Lara Croft	19 FEBRUARY 2002
Ada Wong	05 APRIL 2001
Link	28 JULY 2002

- **TO_CHAR (number)**

  Script:

  SELECT nama,

  TO_CHAR(tinggi_badan) AS tinggi_text

  FROM mahasiswa;

NAMA	TINGGI_TEXT
Socrates	170
Plato	168
Aristotle	172
Friedrich Nietzsche	175
Albert Camus	173
Marcus Aurelius	178
Leonardo da Vinci	176
Pablo Escobar	174
Geralt of Rivia	180
Arthur Morgan	182
Ezio Auditore	177
Kratos	190
Lara Croft	167
Ada Wong	165
Link	169

- **TO_DATE**

  Script:

  SELECT TO_DATE('17-08-2024','DD-MM-YYYY') AS contoh_date
  FROM dual;

```
CONTOH_DA
---------
17-AUG-24
```

d. Fungsi Numerik

- **ABS**

  Script:

  SELECT
  nama,
  berat_badan,
  ABS(berat_badan-70) AS selisih
  FROM mahasiswa;

NAMA	BERAT_BADAN	SELISIH
Socrates	65	5
Plato	63	7
Aristotle	67	3
Friedrich Nietzsche	70	0
Albert Camus	68	2
Marcus Aurelius	74	4
Leonardo da Vinci	72	2
Pablo Escobar	80	10
Geralt of Rivia	78	8
Arthur Morgan	85	15
Ezio Auditore	73	3
Kratos	95	25
Lara Croft	55	15
Ada Wong	53	17
Link	60	10

- **MOD**

  Script:

  SELECT
  nama,
  tinggi_badan,
  MOD(tinggi_badan,5) AS sisa_bagi

FROM mahasiswa;

NAMA	TINGGI_BADAN	SISA_BAGI
Socrates	170	0
Plato	168	3
Aristotle	172	2
Friedrich Nietzsche	175	0
Albert Camus	173	3
Marcus Aurelius	178	3
Leonardo da Vinci	176	1
Pablo Escobar	174	4
Geralt of Rivia	180	0
Arthur Morgan	182	2
Ezio Auditore	177	2
Kratos	190	0
Lara Croft	167	2
Ada Wong	165	0
Link	169	4

- **FLOOR**

  Script:

  SELECT
  nama,
  tinggi_badan,
  FLOOR(tinggi_badan/3) AS floor_value
  FROM mahasiswa;

NAMA	TINGGI_BADAN	FLOOR_VALUE
Socrates	170	56
Plato	168	56
Aristotle	172	57
Friedrich Nietzsche	175	58
Albert Camus	173	57
Marcus Aurelius	178	59
Leonardo da Vinci	176	58
Pablo Escobar	174	58
Geralt of Rivia	180	60
Arthur Morgan	182	60
Ezio Auditore	177	59
Kratos	190	63
Lara Croft	167	55
Ada Wong	165	55
Link	169	56

- **POWER**

  Script:

  SELECT
  nama,
  POWER(tinggi_badan,2) AS pangkat_tinggi
  FROM mahasiswa;

NAMA	PANGKAT_TINGGI
Socrates	28900
Plato	28224
Aristotle	29584
Friedrich Nietzsche	30625
Albert Camus	29929
Marcus Aurelius	31684
Leonardo da Vinci	30976
Pablo Escobar	30276
Geralt of Rivia	32400
Arthur Morgan	33124
Ezio Auditore	31329
Kratos	36100
Lara Croft	27889
Ada Wong	27225
Link	28561

- **ROUND**

  Script:

  SELECT
  nama,
  ROUND(berat_badan/3,2) AS hasil_round
  FROM mahasiswa;

NAMA	HASIL_ROUND
Socrates	21.67
Plato	21
Aristotle	22.33
Friedrich Nietzsche	23.33
Albert Camus	22.67
Marcus Aurelius	24.67
Leonardo da Vinci	24
Pablo Escobar	26.67
Geralt of Rivia	26
Arthur Morgan	28.33
Ezio Auditore	24.33
Kratos	31.67
Lara Croft	18.33
Ada Wong	17.67
Link	20

- **CEIL**

  Script:

  SELECT
  nama,
  CEIL(tinggi_badan/4) AS ceil_value
  FROM mahasiswa;

NAMA	CEIL_VALUE
Socrates	43
Plato	42
Aristotle	43
Friedrich Nietzsche	44
Albert Camus	44
Marcus Aurelius	45
Leonardo da Vinci	44
Pablo Escobar	44
Geralt of Rivia	45
Arthur Morgan	46
Ezio Auditore	45
Kratos	48
Lara Croft	42
Ada Wong	42
Link	43